

Vectors and Vector Spaces

A vector is an ordered list of numbers representing a point or direction in space, e.g. $[2, 5, -1]$. In machine learning, a single data example is almost always represented as a vector: an image's pixels, a document's word counts, or a row of a spreadsheet. A vector space is the set of all vectors of a given dimension, closed under addition and scalar multiplication ? the mathematical structure that lets us add two data points together or scale one by a constant.

Matrices and Matrix Multiplication

A matrix is a rectangular grid of numbers, and can be thought of as a batch of vectors stacked together ? a full dataset of N examples with D features each is naturally an N -by- D matrix. Matrix multiplication combines two matrices by taking dot products of rows and columns; a neural network layer is, at its core, a matrix multiplication followed by a nonlinearity, applied to a batch of input vectors at once.

The Dot Product

The dot product of two vectors multiplies their corresponding entries and sums the results, producing a single number that measures how aligned the two vectors are. A large positive dot product means the vectors point in similar directions; a value near zero means they are roughly perpendicular. This single operation underlies cosine similarity, weighted sums in a neural network neuron, and projecting one vector onto another.